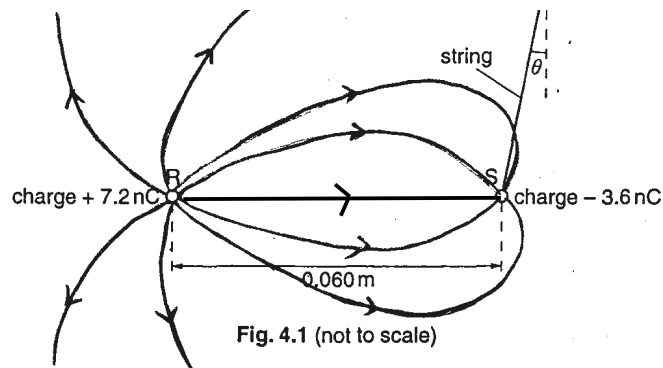


4 (a) (i)



(ii) At mid-point, the electric fields due to R and S point towards S.

$$E = E_R + E_S$$

$$E = \frac{7.2 \times 10^{-9}}{4\pi \times 8.85 \times 10^{-12} \times 0.030^2} + \frac{3.6 \times 10^{-9}}{4\pi \times 8.85 \times 10^{-12} \times 0.030^2}$$

$$E \approx 1.08 \times 10^5 \text{ NC}^{-1}$$

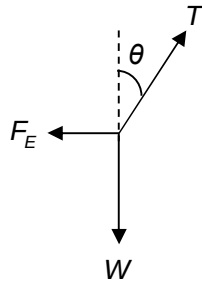
(iii) Using Coulomb's law, we have $F = \frac{Q_R Q_S}{4\pi\epsilon_0 r^2}$ where

$$F = \frac{7.2 \times 10^{-9} \times 3.6 \times 10^{-9}}{4\pi \times 8.85 \times 10^{-12} \times 0.060^2}$$

$$F \approx 6.5 \times 10^{-5} \text{ N (shown)}$$

Note: Candidates are expected to show full substitution of numerical values used.

(b)



Resolving forces horizontally and vertically, we have

$$T \sin \theta = F_E$$

$$T \cos \theta = W$$

$$\text{Hence, } \tan \theta = \frac{F_E}{W}$$

$$\tan \theta = \frac{6.5 \times 10^{-5}}{0.49 \times 10^{-3}}$$

$$\theta \approx 7.56^\circ$$

Therefore, we have

$$T \cos 7.56^\circ = 0.49 \times 10^{-3}$$

$$T \approx 4.94 \times 10^{-4} \text{ N}$$

(c)

